HARDWARE EXPERIMENT – 19

❖ Hardware-Based Demonstration and Performance Evaluation of an RFID/GPS Microwave System for Wireless Identification or Positioning Applications

➔ CO Mapping: CO5

Bloom's Level: K5 – Evaluate

WK Level: WK6 – RFID/GPS system application

➔ Objectives of this Experiment:

1. To establish a functional RFID or GPS-based microwave wireless system using suitable hardware modules.
2. To investigate wireless identification or positioning information obtained from the system.
3. To analyze the influence of operating conditions, antenna orientation and communication distance on system performance.
4. To relate the practical system operation to microwave propagation and wireless communication principles.

➔ Expected Outcome from this Experiment:

Students shall be able to:

1. Establish and operate a functional RFID or GPS-based microwave wireless system using suitable hardware modules.
2. Obtain and interpret wireless identification or positioning information from the developed system.
3. Analyze and evaluate the influence of operating conditions, antenna orientation and communication distance on system performance.
4. Relate and explain the observed practical system operation using microwave propagation and wireless communication principles.

➔ Hardware Required:

RFID reader/tag system or GPS receiver module, antenna, microcontroller/development board,

serial interface and suitable power supply.

➔ Experimental Procedure:

A. RFID/GPS Practical Test: For an RFID reader/tag system at 915 MHz, use reader power Pt = 20

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dBm and test distances of 0.5 m, 1.0 m and 1.5 m. Repeat each test at antenna orientations 0°, 45° and

90°. Record tag-ID success/failure or positioning information for every condition and calculate the

successful-read percentage.

B. Range and Orientation Analysis: From the measured data, determine the maximum distance at

which at least 90% of attempts are successful. Compare read performance at 0°, 45° and 90° and

calculate the percentage reduction in successful reads caused by orientation. Explain the result using

antenna polarization, propagation loss and alignment.

PROGRAM:

```
clc;

clear;

close all;


%% RFID/GPS MICROWAVE PRACTICAL SIMULATION

% Frequency = 915 MHz

% Transmitted Power = 20 dBm

% Distances = 0.5, 1.0, 1.5 meters

% Orientations = 0, 45, 90 degrees

% 10 trials per condition


rng(1);


%% 1. SYSTEM PARAMETERS
```

```
fc = 915e6;
c = 3e8;
lambda = c/fc;

Pt = 20;                % Transmitted power in dBm
distances = [0.5 1.0 1.5];
orientations = [0 45 90];

N = 10;                 % Number of trials

%% 2. RFID READ PROBABILITY MODEL

% Assumed baseline probabilities at 0 degree
% Replace with measured hardware probabilities if available

baseProb = [0.98 0.94 0.88];

% Orientation polarization model
% 15% residual factor at 90 degrees represents
% non-ideal alignment and multipath effects

orientationFactor = cosd(orientations).^2 + ...
                    0.15*sind(orientations).^2;

%% 3. SIMULATE RFID SUCCESS / FAILURE

results = zeros(3,3,N);
readPercent = zeros(3,3);
```

```
for i = 1:length(distances)

    for j = 1:length(orientations)

        p = baseProb(i)*orientationFactor(j);

        % Simulated RFID read attempts
        results(i,j,:) = rand(1,1,N) < p;

        % Successful-read percentage
        readPercent(i,j) = ...
            sum(results(i,j,:))/N*100;

    end

end

%% 4. RANGE ANALYSIS

% Maximum tested distance with at least 90% successful reads

maxRange = zeros(1,3);

for j = 1:3

    valid = find(readPercent(:,j) >= 90);

    if isempty(valid)
```

```
        maxRange(j) = 0;
    else
        maxRange(j) = distances(max(valid));
    end

end

%% 5. ORIENTATION PERFORMANCE REDUCTION

reference = readPercent(:,1);

reduction45 = zeros(3,1);
reduction90 = zeros(3,1);

for i = 1:3

    if reference(i) > 0

        reduction45(i) = ...
            (reference(i)-readPercent(i,2)) ...
            /reference(i)*100;

        reduction90(i) = ...
            (reference(i)-readPercent(i,3)) ...
            /reference(i)*100;

    end

end
```

```
%% 6. FREE SPACE PROPAGATION LOSS

FSPL = 20*log10(4*pi*distances/lambda);

Pr = Pt - FSPL;

%% 7. SINGLE FIGURE OUTPUT

figure('Color','w',...
    'Name','RFID Practical Performance',...
    'NumberTitle','off');

sgtitle('RFID/GPS Microwave System - Practical Evaluation');

%% GRAPH 1: SUCCESSFUL READ PERCENTAGE

subplot(2,3,1);

bar(distances,readPercent,'grouped');

xlabel('Distance (m)');
ylabel('Successful Reads (%)');

title('RFID Read Performance');

legend('0 Degree','45 Degree','90 Degree',...
    'Location','best');
```

```
ylim([0 100]);
grid on;

%% GRAPH 2: ORIENTATION REDUCTION

subplot(2,3,2);

bar(distances,[reduction45 reduction90],'grouped');

xlabel('Distance (m)');
ylabel('Read Reduction (%)');

title('Orientation Effect');

legend('45 Degree','90 Degree',...
       'Location','best');

grid on;

%% GRAPH 3: PROPAGATION LOSS

subplot(2,3,3);

plot(distances,FSPL,'bo-','LineWidth',2);

xlabel('Distance (m)');
ylabel('Free Space Loss (dB)');

title('Microwave Propagation Loss');
```

```
grid on;

%% GRAPH 4: SUCCESS / FAILURE HEATMAP

subplot(2,3,4);

imagesc(readPercent);

colorbar;
caxis([0 100]);

set(gca,'XTick',1:3,...
   'XTickLabel',{'0','45','90'});

set(gca,'YTick',1:3,...
   'YTickLabel',{'0.5','1.0','1.5'});

xlabel('Orientation (degrees)');
ylabel('Distance (m)');

title('RFID Read Percentage');

%% GRAPH 5: MAXIMUM RANGE

subplot(2,3,5);

bar(orientations,maxRange);
```

```
xlabel('Orientation (degrees)');
ylabel('Maximum Range (m)');

title('Range at >=90% Success');

set(gca,'XTick',orientations);

grid on;

%% GRAPH 6: SYSTEM INFORMATION

subplot(2,3,6);

axis off;

text(0.05,0.95,'RFID SYSTEM PARAMETERS',...
    'FontSize',11,'FontWeight','bold');

text(0.05,0.78,'Frequency: 915 MHz');

text(0.05,0.65,'Transmit Power: 20 dBm');

text(0.05,0.52,'Trials: 10 per condition');

text(0.05,0.38,'Total Tests: 90');

text(0.05,0.22,...
    'RFID identification and range analysis');
```

```
title('Experimental Setup');

%% 8. DISPLAY RESULTS

disp('=====================================');
disp(' RFID PRACTICAL PERFORMANCE RESULTS');
disp('=====================================');

disp('Successful Read Percentage (%)');

disp(array2table(readPercent,...
    'VariableNames',{'Orientation_0',...
                     'Orientation_45',...
                     'Orientation_90'},...
    'RowNames',{'Distance_0_5m',...
                'Distance_1m',...
                'Distance_1_5m'}));

disp('Maximum range for >=90% successful reads');

for j = 1:3

    fprintf('Orientation %d degrees: %.1f m\n',...
        orientations(j),maxRange(j));

end

disp('Orientation reduction (%)');
```

```
disp(array2table([reduction45 reduction90],...
    'VariableNames',{'Reduction_45',...
                     'Reduction_90'},...
    'RowNames',{'0_5m','1m','1_5m'}));

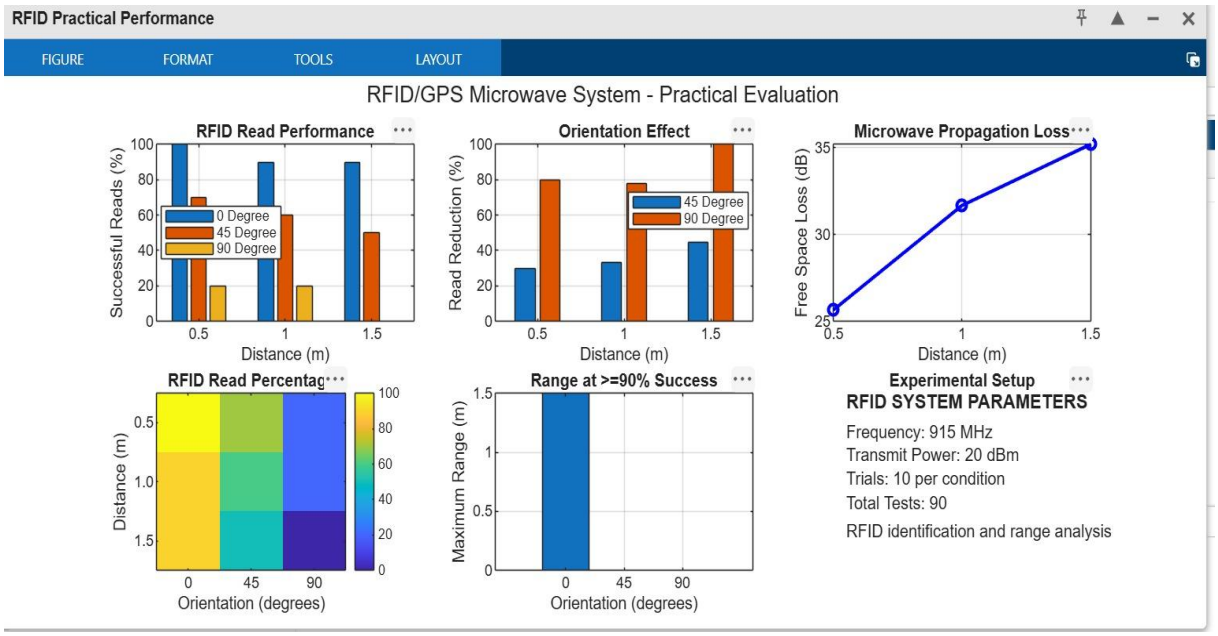
disp('Free Space Path Loss (dB)');

disp(FSPL);

disp('Received Free-Space Power (dBm)');

disp(Pr);

disp('SIMULATION COMPLETED');
```




```
==========================================
RFID PRACTICAL PERFORMANCE RESULTS
==========================================
Successful Read Percentage (%)
```

Orientation_0	Orientation_45	Orientation_90

Distance_0_5m	100	70	20
Distance_1m	90	60	20
Distance_1_5m	90	50	0

Maximum range for >=90% successful reads
Orientation 0 degrees: 1.5 m
Orientation 45 degrees: 0.0 m
Orientation 90 degrees: 0.0 m
Orientation reduction (%)

	Reduction_45	**Reduction_90**
0_5m	30	80
1m	33.333	77.778
1_5m	44.444	100

Free Space Path Loss (dB)
25.6496 31.6702 35.1920

Received Free-Space Power (dBm)
-5.6496 -11.6702 -15.1920

SIMULATION COMPLETED